Multicue Contrastive Self-Supervised Learning for Change Detection in Remote Sensing

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***Abstract*— Contrastive self-supervised learning (CSSL) is a promising method for extracting effective features from unlabeled data. It performs well in image-level tasks, such as image classification and retrieval. However, the existing CSSL methods are not suitable for pixel-level tasks, for example, change detection (CD), since they ignore the correlation between local patches or pixels. In this article, we first propose a multicue CSSL (MC-CSSL) method to derive dense features for CD. Besides data augmentation, the MC-CSSL takes advantage of more cues based on the semantic meaning and temporal correlation of local patches. Specifically, the positive pair is built from local patches with similar semantic meanings or temporal ones with the same geographic location. The assumption is that local patches belonging to the same kind of land-covering tend to share similar features. Second, the affinity matrix is truncated and introduced to extract change information between two temporal patches obtained from different types of sensors. As a result, some initial unchanged pixels are selected to serve as the supervision for mapping the dense features into a consistent space. Based on the distance between all bitemporal pixels in the consistent space, a difference image (DI) is generated and more unchanged pixels can be available. The dense feature mapping and unchanged pixel updating proceed alternately. The proposed CD method is evaluated in both homogeneous and heterogeneous cases, and the experimental results demonstrate its effectiveness and priority after comparison with some existing state-of-the-art methods. The source code will be available at https://github.com/Yang202308/ChangeDetection_CSSL.**

***Index Terms*— Change detection (CD), contrastive self-supervised learning (CSSL), dense features, feature matching, local self-similarity descriptor, remote sensing.**

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I. Introduction

REMOTE-sensing images (RSIs) can provide rich information about land-coverings through satellites or aircraft [1], [2], [3], [4]. Change detection (CD) is a process of monitoring the dynamic changes of the land-coverings by analyzing two (or more) images acquired in the same geographical location but at different times. The applications of CD in remote sensing include urban development [5], deforestation [6], disaster assessment [7], and so on. Various CD approaches have been developed in synthetic aperture radar (SAR) images and multispectral/hyperspectral images [8].

Generally, two temporal RSIs are acquired by the same type of sensors, which is named homogeneous case. There have already existed many approaches to deal with the CD problem in this case, where two coregistered input images can be compared directly to generate a difference image (DI). The ratio operator is effective in SAR images, while the spectral change vector is suitable for CD in multispectral images. To identify changes from the DI, common methods include thresholding [9], clustering [10], statistical modeling [11], change vector analysis (CVA) [12], support vector machines (SVMs) [13], deep neural networks (DNNs) [14], [15], [16], [17], [18], [19], and so on. One kind of DNN-based CD method distinguishes changed areas and unchanged ones with the last layer of the networks, without generating DIs [15], [16], [19]. The supervision was derived from the pseudo-labels of reliable samples with high confidence to be correctly preclassified. Some research focused on transferring pretrained deep features from another task or dataset with labels to CD in remote sensing [17]. For example, the backbone VGG-16 of the feature difference convolutional neural network (CNN) (FDCNN) [18] was pretrained on aerial image data (AID) with scene classification tasks.

CD in heterogeneous cases is a more interesting and challenging problem, in which a pair of images are obtained from different types of sensors [20]. It offers a much greater degree of flexibility in the formation of the input image pair. For example, an optical image of the studied area is available before the disaster (e.g., earthquake and flood). To assess the disaster quickly, one SAR image is more likely to be acquired since it is not sensitive to sunlight and atmospheric conditions. However, the imaging mechanism of SAR sensors is different from that of optical ones. The SAR image content indicates

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the amount of backscatter from the studied area, while the appearance of ground objects in optical images depends on the spectral reflectance signatures. Therefore, different characteristics of the object are recorded in two heterogeneous images, making a direct comparison impossible.

To tackle this challenging problem, two heterogeneous images are mapped into the same observation space [21] or aiming at generating a similar probability density function [22]. In [23], SAR images were transformed with logarithmic operation to get similar statistical distribution properties as the optical ones. In [21], homogeneous pixel transformation (HPT) was proposed to transfer each pixel of one image to the observation space of the other image using K-nearest neighbors technique. After transformation, two heterogeneous images are comparable. Cycle-consistent and adversarial networks [24], [25] were also employed for image translation in remote sensing to generate DIs for CD tasks.

Though acquired by different types of sensors, the land-coverings from two heterogeneous images are identical in unchanged areas. Given this reality, the image structure or representations are assumed to be similar between a pair of unchanged patches. Sun et al. [26] proposed an unsupervised CD framework based on the assumption that patch similarity graph structure should remain consistent between two temporal images. In [27], the difference map was derived by a semisupervised pseudo-Siamese network developed to learn low- and deep-level features separately. The key issue is to learn discriminative image representations from two heterogeneous images and then align them into a common space. Hence, unsupervised learning of abstract and perceptual features has become an essential step for CD in heterogeneous cases. Since CD is a pixel-level prediction task, the existing DNN-based CD methods used generative models to learn dense features, such as AE-based [23], [25], [28], [29] and DBN [30]. However, these generative models are mainly trained with reconstruction error, making them focus on image details but ignore the discriminative information and the correlation between pixels.

To this end, we propose a multicue contrastive self-supervised learning (MC-CSSL) method to learn dense features for pixel-by-pixel predicting tasks, especially for CD. Specifically, the MC-CSSL model is built on transformational, temporal, and semantic cues. It takes advantage of the semantic meaning and temporal correlation of local patches, which provides more cues than data augmentation. On the one hand, the input images are segmented into several classes. Generally, local patches belonging to the same class of land-covering are prone to share similar features. On the other hand, a pair of temporal patches with the same geographic location has a large possibility of containing the same kind of land-covering. With MC-CSSL, the dense features are learned to be discriminative by making local patches of the same class close and those from different classes distant in the embedding space. In addition, the truncated affinity matrix is introduced and developed to extract change information for a pair of heterogeneous images. After comparison, numerous reliable unchanged pixels are selected to provide the supervision information for dense feature mapping and matching processes. In the consistent space, a DI can be generated and more unchanged pixels are available. The dense feature mapping and unchanged pixel updating proceed alternately, leading to a fully unsupervised CD method. The proposed CD method achieves superior experimental results on five datasets and its effectiveness is demonstrated in both homogeneous and heterogeneous cases.

There are three contributions as follows.

1) We propose an MC-CSSL method to derive dense features for CD tasks. The discriminability of dense features is ensured for CD in unlabeled RSIs.
2) The affinity matrix is truncated and introduced to extract change information between two heterogeneous temporal images. It makes reliable initial unchanged pixels available to the feature mapping process.
3) On five homogeneous or heterogeneous datasets, the proposed CD method achieves the best or comparable CD results with the state-of-the-art methods. It demonstrates the effectiveness of the proposed CD method.

The rest of this article is organized as follows. Section II reviews the existing self-supervised learning (SSL) and CSSL methods in both natural images and RSIs. Section III illustrates the proposed MC-CSSL method that takes advantage of transformational, semantic, and temporal cues to learn dense features for CD tasks. Section IV describes the local self-similarity descriptor and the proposed CD method. Section V presents the experimental results and analysis in both homogeneous and heterogeneous cases. Section VI concludes this article.

II. Related Work

In recent years, SSL [31], [32] has delivered breakthrough results in learning visual features from unlabeled data. Various pretext tasks are designed in SSL to provide cues for pretraining networks, such as colorizing images or solving jigsaw puzzles [31]. The supervision for SSL-based methods is generated automatically. In remote sensing, Zhang et al. [33] proposed a rotation awareness-based self-supervised deep model to assist the mid-level feature extraction for SAR target recognition. In [34], the representation learning ability was evaluated on three pretext tasks in SSL for RSI scene classification, including image inpainting, relative position predicting, and contrastive learning. SSL methods usually extract global features with low resolution that are beneficial to image-level prediction tasks.

CSSL [35], [36] is a special case of SSL, which learns image representations by making positive pairs close to each other and negative pairs distant in an embedding space. In [35], CSSL was explained from the perspective of mutual information maximization between latent representations. He et al. [36] considered contrastive learning as the problem of dictionary look-up and built a large dictionary with a queue. Some deep models defined the pretext task by predicting the future in latent space, for example, contrastive predictive coding (CPC) [37]. In CSSL, one negative sample is not enough and therefore the triplet loss was generalized to multiclass N-pair loss to enable $N-1$ negative samples [38]. Other models [36] used multiple views of a shared context to generate positive pairs. Based on several data augmentations, a simple

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framework SimCLR [39] was presented for visual feature learning in natural images.

CSSL has attracted great attention in RSI interpretation with transfer learning or fine-tuning [40], [41], [42], [43], [44]. For RSI classification, contrastive multiview coding (CMC) was used for pretraining with several RS datasets [40]. It yielded better results than that pretrained with natural images. In [41], the anchor patch and its neighboring patches served as positive pairs to get smoothed representations. After pretraining with CSSL, very few labeled samples were used to train a classifier with one or two fully connected (FC) layers in the fine-tuning stage [43], [44]. It has also achieved promising results in CD between heterogeneous images. In [45], seasonal contrast was proposed to learn image representations, especially in remote sensing, taking advantage of time and position invariance. Without supervision, two branches of the pseudo-Siamese network can be trained in a contrastive way [46], where the positive pair is derived from bitemporal images of the same scene. In [47], temporal consistency and location information are considered in constructing positive pairs and negative pairs in SSL for multisensor CD.

Compared with the above CSSL methods, more cues are used in the proposed MC-CSSL method to learn dense features for local patches, including image transformation, temporal consistency, and semantic information. Consequently, the learned features can distinguish different categories of land-covering. It provides a new paradigm for analyzing multitemporal RSIs.

III. MC-CSSL

A. Background

In existing CSSL, positive pairs are built by sampling different transformations of the same image, while negative pairs are built from different images as shown in Fig. 1(a). Generally, some image transformations used in data augmentation are considered. The image representations are learned by making positive pairs similar and negative pairs dissimilar in an embedding space. In the embedding space, several versions of the contrastive loss function are used to make positive pairs close together and negative pairs distant. The *NT-Xent* [39] is used in this article, which is defined as

$$l(\tilde{u},\hat{u}) = -\log\frac{\exp(\text{sim}(g(f(\tilde{u})),g(f(\hat{u})))/\tau)}{\sum_{v\in\Lambda^-}\exp(\text{sim}(g(f(\tilde{u})),g(f(v)))/\tau)} \quad (1)$$

where $\tilde{u}$ and $\hat{u}$ are a positive pair. They are different transformations of the same image u. Λ^- contains several negative samples for $\tilde{u}$ and the negative sample v is another image. g is the projection head with a shallow architecture. It is used for mapping the image features extracted by f into an embedding space, in which the contrastive loss is computed. $\text{sim}(g(f(\tilde{u})), g(f(\hat{u}))$ represents the cosine similarity between two vectors $g(f(\tilde{u}))$ and $g(f(\hat{u}))$. It is the dot product between two normalized vectors. τ is a temperature hyperparameter scaling the similarity scores.

To minimize the contrastive loss $l(\tilde{u},\hat{u})$, the similarity between a positive pair $\tilde{u}$ and $\hat{u}$ is maximized. In the meantime, similarity between several negative pairs $\tilde{u}$ and v is minimized, $v \in \Lambda^-$. After pretraining, the feature extractor f is reused for downstream tasks, while the projection head g is discarded. The learned global representations are transferred to different downstream tasks or fine-tuned on other datasets with limited labels. CSSL has improved the performance of DNNs significantly when a large amount of unlabeled data is available.

B. Dense Features by MC-CSSL

The visual features extracted by f after pretraining are usually image-level since the CSSL can be regarded as instance discrimination. It means that each image can be considered as one class and CSSL methods identify it from other classes though under different transformations, as shown in Fig. 1(a). However, the global and image-level features are not enough to pixel-by-pixel predict tasks due to a lack of spatial information. To learn dense features, each local patch is difficult to consider as one instance because of the high correlation with other patches, especially with those cropped from the same image.

To learn dense features, we propose MC-CSSL that considers several local patches with similar semantic meaning as one class. MC-CSSL takes advantage of the cues from semantic meaning and the temporal correlation as well as the transformational cue. In the following, we will illustrate how to learn dense features with these cues in detail.

1) Transformational Cue: According to SimCLR [39], data augmentation can be grouped into spatial- and appearance-based transformations. The former usually includes crop, resize, flip, rotation, cutout, and so on. The latter involves color distortion, blur, Gaussian noise, and so on. The pretext tasks in existing CSSL are defined with those spatial or appearance transformations. In this article, we mainly focus on the image appearance-based transformations to learn dense features from local patches. As shown in Fig. 1(b), a positive pair $\tilde{u}$ and $\hat{u}$ is generated by

$$\begin{aligned}\tilde{u} &= \tilde{t}\left(u^{\boldsymbol{X}}_{m,n}\right)\\ \hat{u} &= \hat{t}\left(u^{\boldsymbol{X}}_{m,n}\right)\end{aligned} \quad (2)$$

where $\tilde{t},\hat{t} \in \Phi$ and Φ is the set of image appearance-based transformations. $u^{\boldsymbol{X}}_{m,n}$ is a local patch centered at (m,n) from image $\boldsymbol{X}$.

In remote sensing, image quality has quite an impact on the performance of many tasks, such as representation learning, object recognition, and CD. Especially, SAR images are prone to be contaminated by speckle noise, leading to large false alarms. The product model has been demonstrated to be valid for SAR data. The statistical model is given by $I_s = T_s\nu_s$, where ν_s indicates the speckle noise. T_s is the intensity of SAR images before noise contamination, while I_s is the observation. In general, it is assumed that the multilook speckle noise follows Gamma distribution, $\nu_s \sim \Gamma(L, L^{-1})$. L is the number of looks. The noises that affect the optical images are Gaussian distributed, for example, $\nu_o \sim \mathcal{N}(0,\sigma^2)$. Accordingly, we obtain $I_o = T_o + \nu_o$, where I_o and T_o are optical images with and without Gaussian noises. Based on the corruption models in SAR and optical images, image transformation by adding noise with different levels is used in MC-CSSL for dense feature learning.

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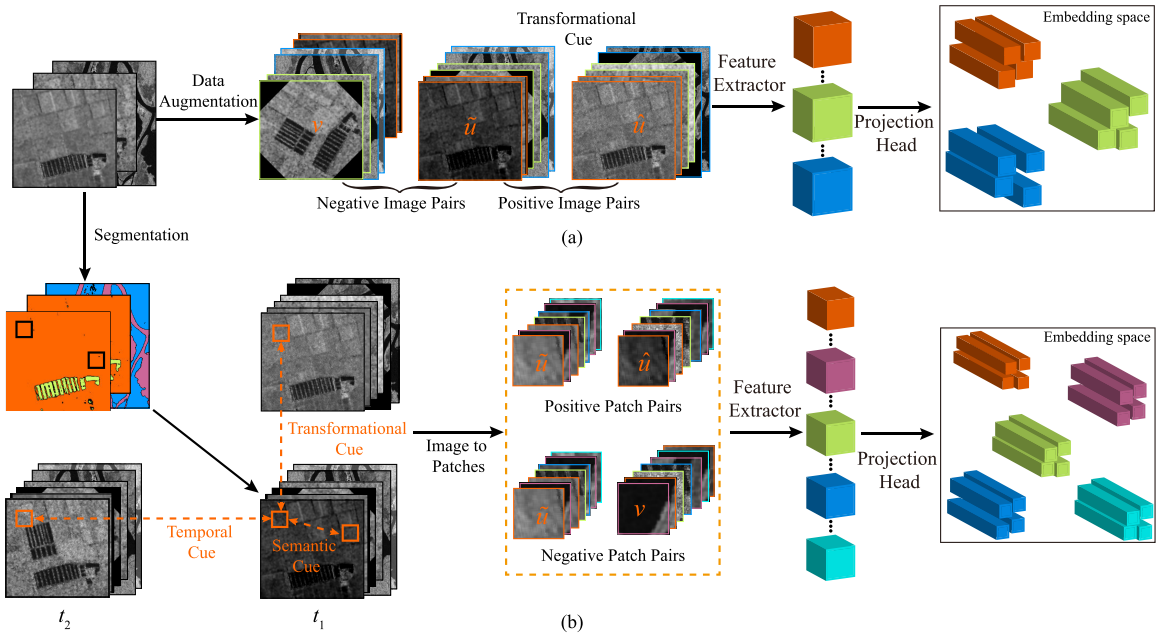


Fig. 1. (a) In the existing CSSL methods, global features are learned based on the transformational cue. In the embedding space, the vectors from different transformations of the same image are encouraged to be close and those from different images are forced to be distant. (b) In MC-CSSL, dense features are learned, taking advantage of the semantic and temporal cues besides the transformational cue. In the embedding space for dense features, the vectors from local patches belonging to the same class are encouraged to be close and those from different classes are distant.

2) Temporal Cue: In remote sensing, it is possible to derive a sequence of temporal images covering the same geographical location but acquired at different times. The temporal consistency between those images can also help us learn the image features in the proposed MC-CSSL.

In CD tasks, two RSIs $\boldsymbol{X}$ and $\boldsymbol{Y}$ are considered with size of $H \times W \times C_1$ and $H \times W \times C_2$, respectively. They share the same spatial size but with different channels. For SAR images, $C_i = 1$, where $i = 1, 2$. For optical images, C_i is equal to the number of bands. Two temporal images have been geometrically registered and radiometrically corrected, which makes it possible to detect the changes in land-coverings in the same geographical location. The positive pair can be built from two local patches $\tilde{u}$ and $\hat{u}$ by

$$\begin{aligned} \tilde{u} &= \tilde{t}\left(u^{\boldsymbol{X}}_{m,n}\right) \\ \hat{u} &= \hat{t}\left(u^{\boldsymbol{Y}}_{m,n}\right) \end{aligned} \tag{3}$$

where $u^{\boldsymbol{X}}_{m,n}$ and $u^{\boldsymbol{Y}}_{m,n}$ are local patches centered at (m, n) from image $\boldsymbol{X}$ and image $\boldsymbol{Y}$, respectively. In the embedding space, two temporal correlated patches are also encouraged to be close. In Fig. 1(b), two local patches correlated in terms of "Temporal Cue" are cropped from the same location in bitemporal images acquired at t_1 and t_2.

3) Semantic Cue: To extract dense features, the proposed MC-CSSL considers each local patch as an instance. However, there are large numbers of local patches that belong to the same type of land-coverings. These patches usually share a similar appearance, for example, farmland areas of the input image shown in Fig. 1. It is difficult and unreasonable to make them distant in the embedding space. On the contrary, they are expected to be close.

To consider the semantic information in CSSL, each input image is segmented into several clusters in an unsupervised manner. Given an RSI $\boldsymbol{X}$, $\boldsymbol{M}^{\boldsymbol{X}}$ is the segmentation map with clustering. The local patches that belong to the same cluster are used to build positive pairs. For example, two patches $\tilde{t}(u^{\boldsymbol{X}}_{m,n})$ and $\tilde{t}(u^{\boldsymbol{X}}_{m',n'})$ correlated in terms of "Semantic Cue" in Fig. 1(b) belong to the category of farmland, $\boldsymbol{M}^{\boldsymbol{X}}(m, n) = \boldsymbol{M}^{\boldsymbol{X}}(m', n')$. After mapped into the embedding space, they are forced to be close. More positive pairs can be generated using semantically similar patches cropped from the same input image.

It is worth noting that spatial correlation is also available when the semantic cue is applied in dense feature learning. The fact is that the neighboring local patches are usually grouped into the same cluster. Compared with the transformational cue in existing CSSL methods, the semantic cue takes advantage of the correlation between different patches and helps learn more informative features.

Since multiple cues are available in MC-CSSL, they can be used one by one. In other words, several pretext tasks are designed and each task relates to one cue. Those pretext tasks are then trained to learn the dense features. We can also design one pretext task using all the mentioned cues at the same time. For example, we can build a positive pair with $\hat{t}(u^{\boldsymbol{X}}_{m,n})$ and $\tilde{t}(u^{\boldsymbol{Y}}_{m',n'})$, where $\boldsymbol{M}^{\boldsymbol{X}}(m, n) = \boldsymbol{M}^{\boldsymbol{X}}(m', n')$. Based on the semantic cue, the negative pair is built from the patches that belong to different classes, as shown in Fig. 1(b). It is reasonable to make them distant in the embedding space.

IV. Proposed CD Method

Fig. 2 shows the pipeline of the proposed CD method used in both homogeneous and heterogeneous cases. At the first stage, the network is pretrained to generate dense features based on MC-CSSL as shown in Fig. 2(a). After pretraining, the projection head is discarded while the feature extractor is reused at the second stage shown in Fig. 2(b). The dense

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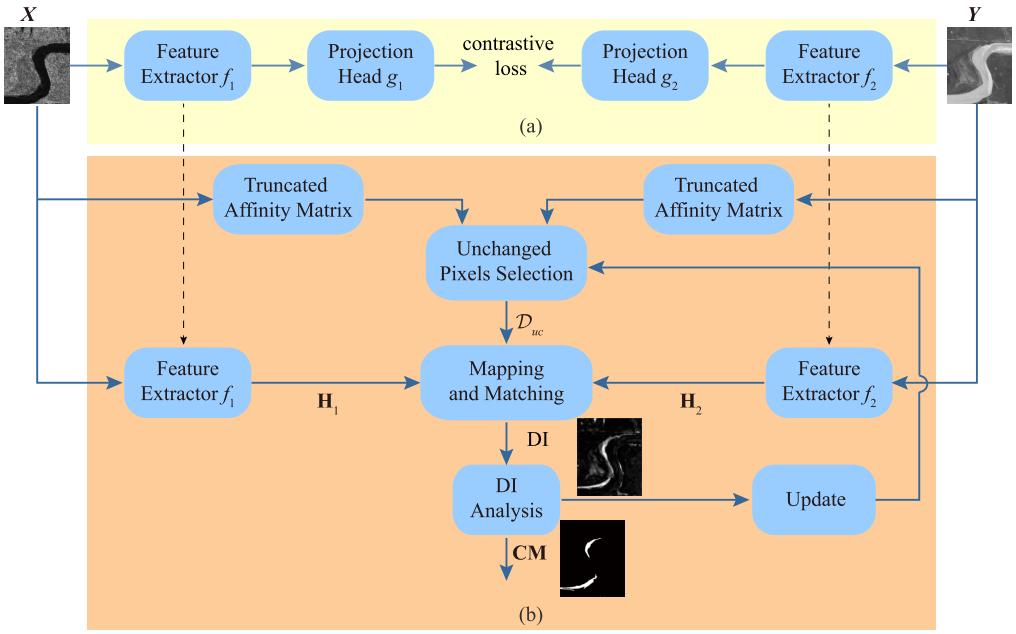


Fig. 2. Pipeline of the proposed CD method in homogeneous and heterogeneous cases. It consists of two stages. (a) In the first stage, the network is pretrained to generate dense features for each of the input images, based on MC-CSSL. After pretraining, the feature extractor is reused, and the projection head is discarded. In the second stage, dense features or local self-similarity descriptors are used in selecting some initial unchanged pixels. After mapping the dense features in the same space, the unchanged pixels are matched exactly. The DI generated by comparing the dense features in the same space can help select more unchanged pixels. The dense feature mapping and unchanged pixel updating proceed alternately. (b) CD based on feature mapping and matching.

features extracted from two input images should be mapped into the same feature space to be comparable. In this space, the paired pixels are matched exactly for pixels identified as the unchanged class. To derive the consistent space, the dense features from one input image can be fixed. Then, those from the other image are mapped to the consistent space.

Algorithm 1 Proposed CD Method

Input:

Two registered remote sensed images $\boldsymbol{X}$ and $\boldsymbol{Y}$, dense features $\mathbf{H}_1$ and $\mathbf{H}_2$ after pretraining, parameters θ_1 and θ_2, local self-similarity descriptors for $\boldsymbol{X}$ and $\boldsymbol{Y}$ based on Eq. (4) and (5);

1: Initialization: The initial unchanged pixels are selected after comparison between dense features or local self-similarity descriptors;
2: Based on the unchanged pixels, minimizing the loss function $\mathcal{L}_c$ in Eq. (6), and updating θ_2 (or θ_1) based on back-propagation algorithm;
3: Fixing θ_1 and θ_2, generating a DI via Eq. (7);
4: Updating the unchanged pixels based on the DI;
5: Repeating steps 2, 3, and 4 until the loss function in Eq. (6) remains stable;
6: Analyzing the DI to yield a CD map;

Output:

The CD map: **CM**.

The proposed CD method is illustrated in Algorithm 1. At first, dense features or local self-similarity descriptors are compared to select some initial unchanged pixels. In homogeneous cases, the dense features $\mathbf{H}_1 = f(\boldsymbol{X}, \theta_1)$ and $\mathbf{H}_2 = f(\boldsymbol{Y}, \theta_2)$ after pretraining can be compared directly since two input images with the same imaging modality can share one DNN-based feature extractor, $\theta_1 = \theta_2$. In heterogeneous cases, the local self-similarity descriptor is introduced to extract the change information. Second, the paired dense features are mapped into a consistent space and matched exactly for initial unchanged pixels. Suppose that one feature extractor f with θ_1 is frozen and the other feature extractor f with θ_2 is updated based on the propagation algorithm. Third, both θ_1 and θ_2 are fixed and the dense features for all pixels are compared to generate a DI. Fourth, the unchanged pixels are updated based on the DI. The dense features and the unchanged areas are optimized alternately. When the loss function in (6) remains stable, the network stops training and a final DI with strong separability is achieved. At last, the final DI is analyzed to yield the changed detection map **CM**.

A. Local Self-Similarity Descriptors

In homogeneous cases, dense features are comparable since they can be extracted by the feature extractor f with the same network parameters, $\theta_1 = \theta_2$. However, two temporal images in heterogeneous cases demonstrate different image appearance and statistics. Correspondingly, $\mathbf{H}_1$ and $\mathbf{H}_2$ are extracted with two different feature extractors $\theta_1 \neq \theta_2$, which are not comparable. Therefore, a domain-invariant local descriptor is required in heterogeneous cases to derive the initial unchanged pixels.

In this article, the local self-similarity descriptor is introduced to extract change information. The assumption is that two pixels or local patches that share a similar appearance in one domain are also similar in another domain. For example, all pixels in the areas of the river are similar in SAR images as shown in Fig. 3. It can be seen that all pixels that belong

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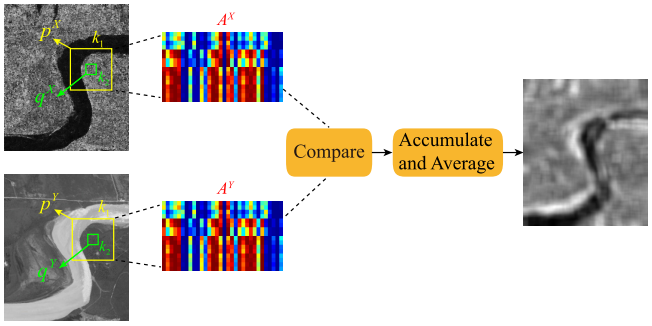


Fig. 3. Element of the truncated affinity matrix, for example, $A^{\boldsymbol{X}} \in \mathbb{R}^{k_1^2 \times k_2^2}$, is the distance between each pixel in one patch and that in the other patch with the same center, for example, $p^{\boldsymbol{X}}$ and $q^{\boldsymbol{X}}$. Two matrices $A^{\boldsymbol{X}}$ and $A^{\boldsymbol{Y}}$ computed from two heterogeneous images are compared to select the initial unchanged pixels.

to the class of river are also similar in the optical images. In [48], the affinity matrix was introduced as the local self-similarity descriptor for comparison between images with different imaging modalities. Specifically, it considered the distance between every two pixels in the local patch, where two pixels with a small geometric distance share a similar appearance. As a result, there is much redundant information in affinity matrices, which also need more computation time.

Fig. 3 illustrates how to compute the local self-similarity with the truncated affinity matrix. At first, two images $\boldsymbol{X}$ and $\boldsymbol{Y}$ are cropped into several overlapping patches with the same size of $k_1 \times k_1$. Let $p^{\boldsymbol{X}}$ and $p^{\boldsymbol{Y}}$ denote a pair of patches from the same location in $\boldsymbol{X}$ and $\boldsymbol{Y}$, respectively. At the same time, a pair of small patches $q^{\boldsymbol{X}}$ and $q^{\boldsymbol{Y}}$ are cropped from $p^{\boldsymbol{X}}$ and $p^{\boldsymbol{Y}}$ with size of $k_2 \times k_2$, respectively. For each input image, for example, $\boldsymbol{X}$, the patch $p^{\boldsymbol{X}}$ is compared with a small patch $q^{\boldsymbol{X}}$ with the same center. The truncated affinity matrix $A^{\boldsymbol{X}}_{m,n}$ between $p^{\boldsymbol{X}}$ and $q^{\boldsymbol{X}}$ with location (m, n) in $\boldsymbol{X}$ is computed by

$$A^{\boldsymbol{X}}_{m,n} = \exp\left\{-\frac{d\left(p^{\boldsymbol{X}}_m, q^{\boldsymbol{X}}_n\right)^2}{h^2}\right\} \tag{4}$$

where $m = 1, \ldots, k_1^2$ and $n = 1, \ldots, k_2^2$. $d(p^{\boldsymbol{X}}_m, q^{\boldsymbol{X}}_n)$ is the distance between the mth pixel in $p^{\boldsymbol{X}}$ and the nth pixel in $q^{\boldsymbol{X}}$. In (4), h is the mean of the distances between $p^{\boldsymbol{X}}_m$ and the centered pixel of $q^{\boldsymbol{X}}$. Correspondingly, the truncated affinity matrix $A^{\boldsymbol{Y}}_{m,n}$ for $\boldsymbol{Y}$ is also computed as

$$A^{\boldsymbol{Y}}_{m,n} = \exp\left\{-\frac{d\left(p^{\boldsymbol{Y}}_m, q^{\boldsymbol{Y}}_n\right)^2}{h^2}\right\}. \tag{5}$$

The distance between a pair of patches $p^{\boldsymbol{X}}$ and $p^{\boldsymbol{Y}}$ is measured by $||A^{\boldsymbol{X}} - A^{\boldsymbol{Y}}||_F$ that is the Frobenius norm of the difference between $A^{\boldsymbol{X}}$ and $A^{\boldsymbol{Y}}$. Then, the value is assigned to each pixel in $p^{\boldsymbol{X}}$ and $p^{\boldsymbol{Y}}$. To get a whole distance map between $\boldsymbol{X}$ and $\boldsymbol{Y}$, the overlapping area between two adjacent patches is averaged. To get the initial unchanged area, M_I pixels with the lowest distance are selected.

The truncated affinity matrix focuses on the distance between a large patch $p^{\boldsymbol{X}}$ and a small patch $q^{\boldsymbol{X}}$ with the same center, which improves the efficiency. In addition, $q^{\boldsymbol{X}}$ and $q^{\boldsymbol{Y}}$ can also be the downsampled version of $p^{\boldsymbol{X}}$ and $p^{\boldsymbol{Y}}$, respectively. The local self-similarity is also efficient with less computation time. The idea was inspired by Shechtman and Irani [49], which is always used in sketch matching and not suitable in regions with less structure and texture information.

B. Mapping and Matching

For the CD problem, a pair of unchanged pixels belong to the same kind of land-covering. Therefore, only the unchanged pixels can be matched exactly. In feature mapping, suppose that parameters θ_1 of the first feature extractor are frozen and the dense features $\mathbf{H}_1$ are fixed. The loss function $\mathcal{L}_c$ of matching $\mathbf{H}_1$ and $\mathbf{H}_2$ for unchanged pixels is defined by

$$\mathcal{L}_c(\theta_2) = \sum_{(m,n)\in\mathcal{D}_{\text{uc}}} \|\mathbf{h}_1(m,n) - \mathbf{h}_2(m,n)\|_2 \tag{6}$$

where $\mathcal{D}_{\text{uc}}$ is the set of M_I unchanged pixels, according to the difference map between two input images $\boldsymbol{X}$ and $\boldsymbol{Y}$. $\mathbf{h}_1(m,n)$ and $\mathbf{h}_2(m,n)$ are feature vectors at position (m,n) in $\boldsymbol{X}$ and $\boldsymbol{Y}$, respectively. It can be seen that only θ_2 is updated in the feature mapping and matching process.

After the loss function $\mathcal{L}_c$ is minimized, the network parameters θ_1 and θ_2 can be fixed at the same time. Then, a DI is computed by

$$\text{DI}(m,n) = \|\mathbf{h}_1(m,n) - \mathbf{h}_2(m,n)\|_2. \tag{7}$$

When $\text{DI}(m,n)$ is high, it indicates that the pixel (m,n) is prone to be changed. On the contrary, it may be an unchanged pixel.

C. Unchanged Pixels Updating

Though $\mathcal{D}_{\text{uc}}$ with M_I unchanged pixels can serve as the supervision in the feature mapping and matching process, it is not enough to get a DI with high-quality for the CD task. As mentioned above, the pixel with a low value in a DI has a large possibility to be unchanged. Therefore, more unchanged pixels can be estimated based on the DI and then added to $\mathcal{D}_{\text{uc}}$.

Specifically, all the pixels are sorted based on the DI after a certain number of iterations. Then, some pixels with the lowest value of DI are considered as unchanged pixels. Since the DI is not satisfied at the beginning of the feature mapping and matching process, only a small number of unchanged pixels are selected and used. After several iterations, more unchanged pixels can be added to $\mathcal{D}_{\text{uc}}$.

To sum up, the dense feature matching in (6) and the unchanged pixels updating can proceed alternately. At first, $\mathcal{D}_{\text{uc}}$ only contains the initial unchanged pixels based on the comparison between self-similarity matrices or pretrained dense features $\mathbf{H}_1$ and $\mathbf{H}_2$ extracted from two input images. Then, $\mathbf{H}_1$ and $\mathbf{H}_2$ are mapped into a consistent space and then matched in the unchanged area. With the DI computed by the distance between $\mathbf{H}_1$ and $\mathbf{H}_2$, $\mathcal{D}_{\text{uc}}$ is updated to contain more unchanged pixels that are beneficial to the feature mapping and matching process.

V. Experiments

In the following, the proposed CD method is evaluated in both the homogeneous and heterogeneous cases. It is

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TABLE I
BITEMPORAL DATASETS DESCRIPTIONS

Case	Data Set	time t_1	Data Source at time t_1	time t_2	Data Source at time t_2	Size
Homogeneous	Farmland	June, 2008	Radarsat-2 (SAR)	June, 2009	Radarsat-2 (SAR)	306× 291
	Ottawa	May 1997	Radarsat (SAR)	Aug. 1997	Radarsat (SAR)	290× 350
Heterogeneous	River	June, 2008	Radarsat-2 (SAR)	Sep. 2010	Google Earth (Optical)	291× 343
	Shuguang Village	June, 2008	Radarsat-2 (SAR)	Sep. 2012	Google Earth (Optical)	921× 593

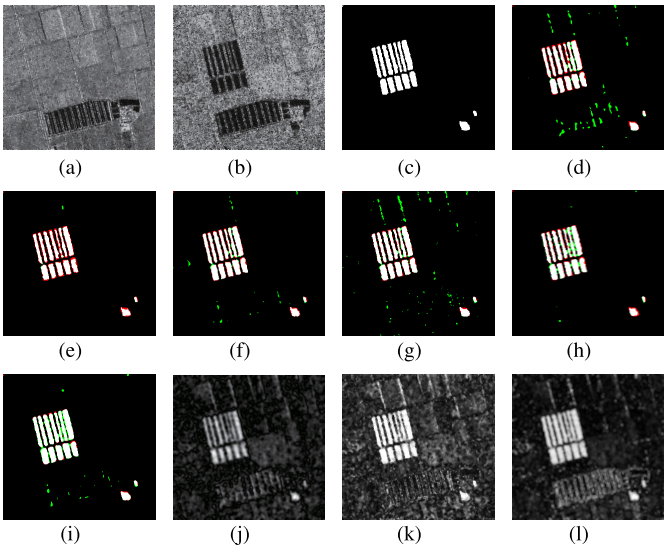


Fig. 4. Farmland dataset and the corresponding CD results. (a) SAR image acquired before the flood. (b) SAR image acquired after the flood. (c) Reference image. The confusion maps by (d) GKI, (e) PCANet, (f) D_RFLICM, (g) AM_HPT, (h) RDN, and (i) proposed. In the confusion maps, White: TP, Black: TN, Green: FP, Red: FN. The DIs of the Farmland dataset by some DI-based CD methods, including (j) GKI, (k) AM_HPT, and (l) proposed.

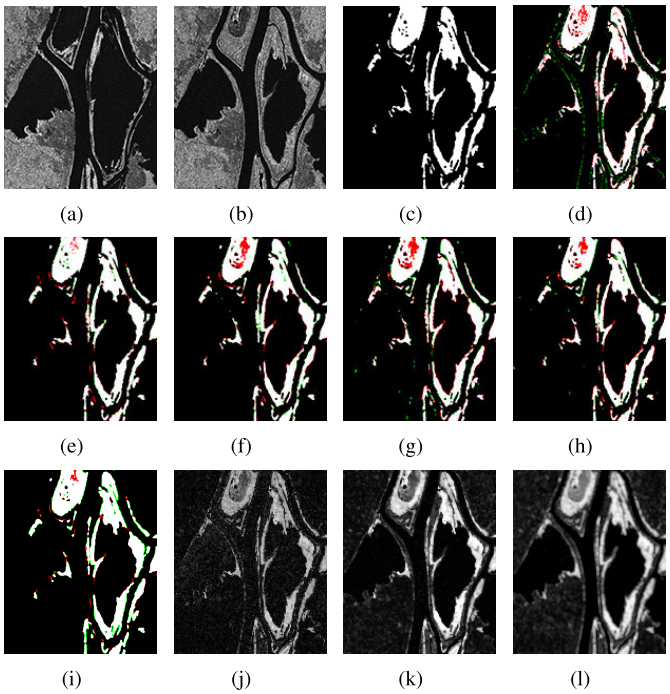


Fig. 5. Ottawa dataset and the corresponding CD results. (a) SAR image taken before the flood. (b) SAR image after the flood. (c) Reference image. The confusion maps by (d) GKI, (e) PCANet, (f) D_RFLICM, (g) AM_HPT, (h) RDN, and (i) proposed. In the confusion maps, White: TP, Black: TN, Green: FP, Red: FN. The DIs of Ottawa dataset by some DI-based CD methods, including (j) GKI, (k) AM_HPT, and (l) proposed.

also compared with various state-of-the-art CD algorithms to demonstrate its effectiveness and priority. In the discussion, some modules are analyzed including the dense features learning, self-similarity matrices, and feature mapping and matching process. Finally, the proposed CD method is also evaluated on multitemporal datasets.

A. Datasets

Table I provides a brief description of four bitemporal datasets, including two homogeneous cases and two heterogeneous cases. The Farmland dataset consists of two SAR images with a size of 306 × 291 and spatial resolution of 8 m, as shown in Fig. 4(a) and (b). They were acquired by Radarsat-2, providing a view of farmland changes in Fig. 4(c) at the region of Yellow River Estuary in China.

The Ottawa dataset was acquired by Radarsat over the city of Ottawa. In Fig. 5(a) and (b), two SAR images with size of 290 × 350 are provided by Defense Research and Development Canada. The spatial resolution for the Ottawa dataset is 10 m. The land-covering changes were recorded before and after the summer flooding.

The River dataset was acquired by different types of sensors. One is a SAR image in Fig. 6(a) by Radardat-2, and the other is an optical image in Fig. 6(b) from Google Earth. Google Earth is powered by satellites, such as Landsat-7, IKONOS, QuickBird, and so on. Two images are with the same size of 291 × 343 and a spatial resolution of 8 m, covering the same region of the Yellow River Estuary.

Shuguang Village dataset consists of one C-band SAR image by Radarsat-2 and one RGB image from Google Earth as shown in Fig. 7(a) and (b), respectively. The resolution is 8 m and the spatial size is 921 × 593. The reference image in Fig. 7(c) refers to the change of land use in the Shuguang Village of the Dongying City in China.

B. Experimental Settings

1) Evaluation Measures: To evaluate the performance of the proposed CD method, some quantitative evaluation indices are used. According to the reference image, FP represents the number of false positives that are unchanged pixels in the reference image but detected as changed ones in the predicted CD maps. FN is the number of false negatives, which are detected as unchanged pixels incorrectly. The overall error OE = FP + FN. The overall accuracy is the percentage of correctly detected pixels (PC), PC = $(\mathrm{TP}+\mathrm{TN})/(N_c+N_u)$. TP and TN represent the true positives and true negatives, respectively. N_c is the number of changed pixels and N_u is

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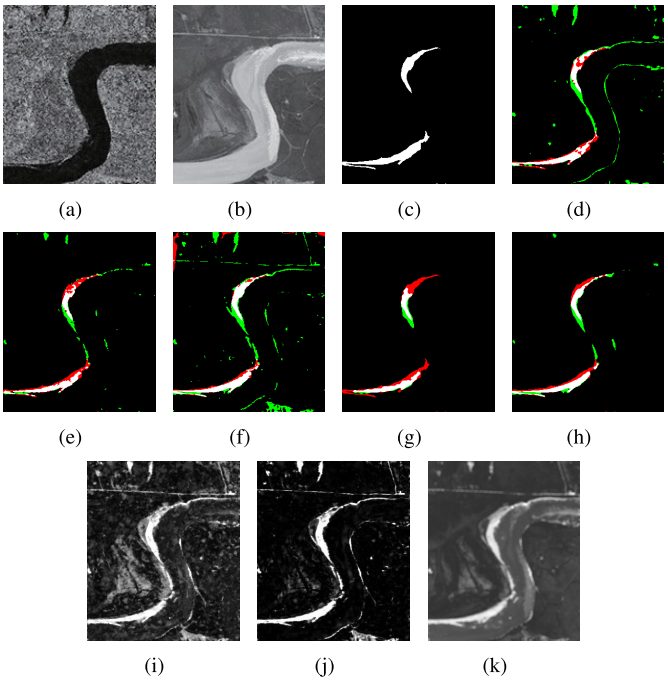


Fig. 6. River dataset and the corresponding CD results. (a) SAR image acquired in June 2008. (b) Optical image acquired in September 2010. (c) Reference image. The confusion maps by (d) PCC, (e) AM_HPT, (f) CA_AE, (g) FPMSMCD, and (h) proposed. In the confusion maps, White: TP, Black: TN, Green: FP, Red: FN. The DIs of the River dataset by some DI-based CD methods, including (i) AM_HPT, (j) CA_AE, and (k) proposed.

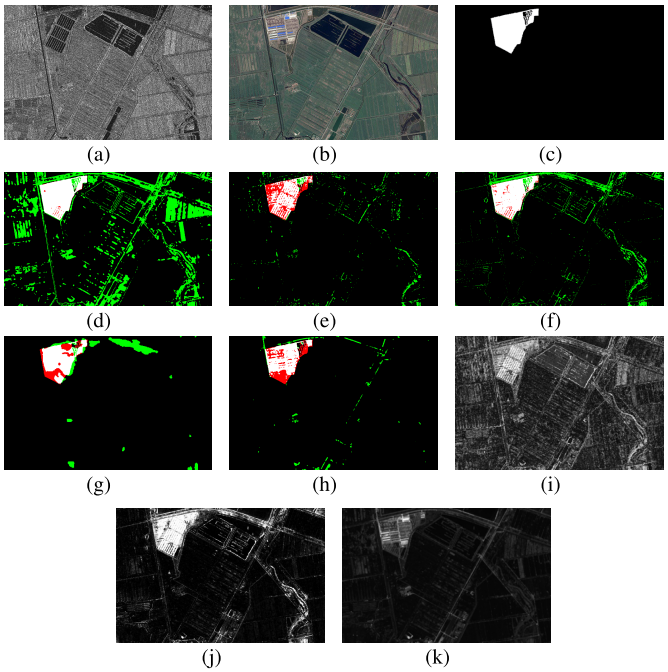


Fig. 7. Shuguang dataset and the corresponding CD results. (a) SAR image acquired in June 2008. (b) Optical image acquired in September 2012. (c) Reference image. The confusion maps by (d) PCC, (e) AM_HPT, (f) CA_AE, (g) MCD, and (h) proposed. In the confusion maps, White: TP, Black: TN, Green: FP, Red: FN. The DIs of the River dataset by some DI-based CD methods, including (i) AM_HPT, (j) CA_AE, and (k) proposed.

that of the unchanged ones in the reference image. Kappa coefficient (KC) is also an important measure that indicates the consistency between the predicted CD maps and the reference image. It is calculated by $\mathrm{KC} = (\mathrm{KC} - \mathrm{PRE})/(1 - \mathrm{PRE})$, where $\mathrm{PRE} = (N_c \cdot (\mathrm{TP} + \mathrm{FP}) + N_u \cdot (\mathrm{TN} + \mathrm{FN}))/(N_c + N_u)^2$.

In addition, we also evaluate the DI generated by various DI-based methods. Following the protocol used in [28], the receiver-operating characteristic (ROC) curve is considered the performance index, which is produced by plotting the TP rate (TPR) versus the FP rate (FPR). At each threshold, TPR is defined by $\mathrm{TPR} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$ and FPR by $\mathrm{FPR} = \mathrm{FP}/(\mathrm{FP} + \mathrm{TN})$. The area under the ROC curve (AUC) is also used to evaluate the DI. An excellent DI has an AUC value near 1.0, as well as an ROC curve with a point in the top-left corner of the ROC space. A high-quality DI makes it easy to distinguish between the unchanged class and the changed one. With AUC and ROC, the DI-based CD methods can be measured directly, no matter which approach is adopted in DI analysis.

2) Implementation Details: In the proposed MC-CSSL, all the SAR patches from four datasets are used to learn dense features. Thus, two temporal images in homogeneous cases share one feature extractor in the pretraining. To pretrain the feature extractor, one pretext task is designed using transformational, temporal, and semantic cues sequentially. The speckle noise is added to the SAR patches to generate the positive pairs. For the temporal cue, MC-CSSL considers the patches with the same location and similar appearance from two homogeneous images to build positive pairs. To leverage the semantic cue, each input image is segmented using a clustering algorithm, for example, fuzzy local information C-means (FLICM). The size of local patches is 7×7 empirically.

In heterogeneous cases, two feature extractors are pretrained by the networks with the same architecture but different parameters. We adopt the network architecture from [50] to design the feature extractor, using three convolutional blocks. Each convolutional block contains a 1×1 convolution layer with 64 filters, a batch normalization, and the rectified linear unit (ReLU) function. The projection head consists of one convolutional layer and one FC layer. The batch size is 512 and the learning rate is 0.001 for dense features pretraining. The network parameters are optimized by a gradient descent algorithm. The feature extractors are pretrained with 100 epochs. For the feature mapping process, the network parameter updating stops when the loss function in (6) stabilizes. To validate the effectiveness of visual feature learning and feature matching process, CD maps are obtained by a thresholding approach based on the final DI.

For self-similarity matrices, k_1 is set to 20 as in [48]. If q^X and q^Y are cropped from the center of p^X and p^Y, k_2 is set to 2 empirically.

3) Comparison Methods: The proposed method is also compared with various CD methods. Some approaches developed for homogeneous cases cannot be used in the CD task in heterogeneous cases with different imaging modalities.

In homogeneous cases, GKI [51] used thresholding for the magnitude of the DI that was assumed to be generalized Gaussian distributed. PCANet [52] performed convolution with PCA filters to extract features for each pixel. In this article, the publicly available code for PCANet was used.

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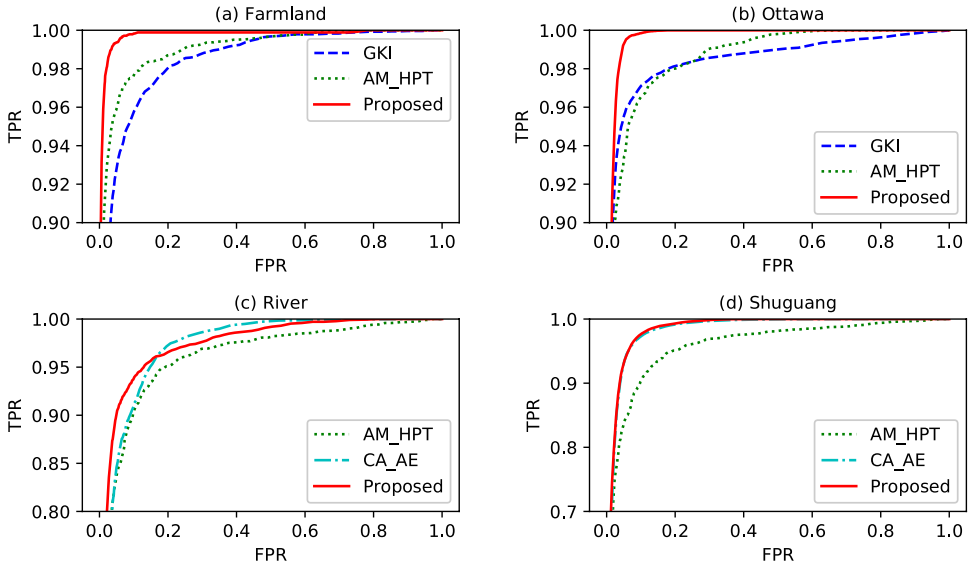


Fig. 8. ROC curve of the DIs by some DI-based CD methods on (a) Farmland, (b) Ottawa, (c) River, and (d) Shuguang datasets.

TABLE II
CD RESULTS ON THE FARMLAND DATASET

Methods	KC	PC(%)	OE	AUC
GKI	0.8005	97.81	1948	0.9834
PCANet	0.8553	98.56	1284	/
D_RFLICM	0.8950	**98.86**	**1013**	/
AM_HPT	0.8561	98.36	1459	0.9901
RDN	0.8842	98.74	1122	/
Proposed	**0.8971**	98.79	1077	**0.9974**

TABLE III
CD RESULTS ON THE OTTAWA DATASET

Methods	KC	PC(%)	OE	AUC
GKI	0.8839	96.94	3103	0.9840
PCANet	**0.9326**	**98.22**	**1807**	/
D_RFLICM	0.9087	97.62	2412	/
AM_HPT	0.8746	96.75	3300	0.9863
RDN	0.9021	97.45	2593	/
Proposed	0.9002	97.28	2758	**0.9957**

In D_RFLICM [53], a preclassification map is generated by RFLICM to select a set of pixels with a high possibility of being classified correctly. Those pixels are used to train a DNN-based CD model. The neighborhood size is set to 5 for D_RFLICM. AM_HPT [48] mainly compared the unchanged pixels of the same location and considered it as a regression problem. For AM_HPT, SAR images are preprocessed by speckle noise reduction and log transformation. RDN [14] is a DNN-based CD method with domain adaptation. It adapted the knowledge about land-cover changes from one domain into another domain.

In heterogeneous cases, PCC is a postclassification comparison method that needs land-cover classification maps [54] with high quality. CA_AE [29] learned comparable features between heterogeneous images with code-aligned AEs. In CA_AE, the patch size is set to 50×50 and the average results of ten times are recorded. FPMSMCD [55] mapped the one image to the imaging modality of the other image. MCD [56] used the Bayesian statistical approach to model the dependencies between different imaging modalities in CD tasks. Default settings are used for FPMSMCD and MCD.

C. Experimental Results

1) Farmland Dataset: Table II lists the CD results of the Farmland dataset, which are achieved by the proposed method as well as other approaches for comparison. Before the CD maps, the GKI, AM_HPT, and the proposed method generate DIs at first that are also evaluated with AUC values. A simple ratio operator was used in GKI, leading to a rough DI with an AUC value of 0.9834. Compared with GKI and AM_HPT, the proposed method uses dense feature matching to generate a high-quality DI with an AUC value of 0.9974 and achieves the highest value for KC of 0.8971. It demonstrates the priority of MC-CSSL in learning informative dense features and the efficiency of the proposed CD methods. The best results are highlighted in bold.

The confusion maps for the Farmland dataset are shown in Fig. 4(d)–(i). In the confusion maps, FP and FN are depicted in green and red, respectively. It is observed that PCANet gets the least FPs but the most FNs. The deep models including D_RFLICM, RDN, and the proposed CD method achieve satisfied results with the hierarchical features. The DI maps for Farmland dataset are also shown in Fig. 4(j)–(l) as well as the ROC curves in Fig. 8(a).

2) Ottawa Dataset: In Table III, the CD results are evaluated quantitatively for the Ottawa dataset. PCANet yields the best KC of 0.9326 and PC of 98.22%, indicating that the predicted CD map is nearly in agreement with the reference image. In PCANet, the final change map was the combination of the classification map from a linear SVM and the preclassification map from fuzzy C-means (FCM). On the Ottawa dataset, D_RFLICM and RDN get similar results, which identify the changes with the last layer of the network.

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TABLE IV
CD RESULTS ON THE RIVER DATASET

Methods	KC	PC(%)	OE	AUC
PCC	0.5186	96.28	3690	/
AM_HPT	0.5825	97.50	2483	0.9568
CA_AE	0.5353	96.03	3963	0.9666
FPMSMCD	0.5985	**97.94**	**2053**	/
Proposed	**0.6368**	97.80	2197	**97.16**

TABLE V
CD RESULTS ON THE SHUGUANG DATASET

Methods	KC	PC(%)	OE	AUC
PCC	0.2733	83.26	91398	/
AM_HPT	0.5705	96.59	18608	0.9646
CA_AE	0.5785	95.03	27144	0.9824
MCD	0.6300	96.74	17805	/
Proposed	**0.6816**	**97.64**	**12881**	**0.9848**

In D_RFLICM and RDN, the supervision from (pseudo-)labels was used. They also used a softmax classifier and much information in the neighborhood, which may be more capable of recognizing changes than the thresholding used in the proposed method. In Table III, the proposed method generates a DI with strong separability and an AUC value of 0.9957.

The ROC curves of the DIs for all datasets are shown in Fig. 8. In Fig. 8(b), it can be seen that a point exists close to the top-left corner of the ROC space. It validates the efficacy of the proposed CD method. The visual CD maps for the Ottawa dataset are shown in Fig. 5, as well as the DIs of some methods. From Fig. 5, PCANet yields the least FPs in the CD map, while the proposed method gets the least FNs. In the DIs by AM_HPT and the proposed method, changed areas are highlighted and the unchanged areas are suppressed.

3) River Dataset: The CD task in heterogeneous cases is challenging since two input images demonstrate different appearance and statistics, as shown in Fig. 6(a) and (b). In Table IV, the quantitative results are given for various CD methods. PCC gets poor CD results due to the accumulated errors of the land-covering classification maps. For CA_AE, it may need more training samples to achieve good results. However, the River dataset only contains a small number of pixels. FPMSMCD and the proposed method achieve better performance with the largest PC of 97.94% and the largest KC of 0.6368, respectively.

The confusion maps are visualized in Fig. 6 for the River dataset. It can be seen that the number of FPs is the largest by PCC and CA_AE. For DIs shown in Fig. 6(i) and (j), some unchanged areas are also highlighted to generate false alarms. The proposed method and FPMSMCD yield the smallest errors on the River dataset. It proves that the dense features of unchanged areas can be matched exactly, even though they are extracted from one SAR image and one optical image.

4) Shuguang Dataset: The CD results on the Shuguang dataset are shown in Table V and Fig. 7. From Table V, it can be seen that AM_HPT gets a small value for AUC, indicating that the DI is poor. The separability of DI is weak to correspondingly yield a small value for KC. As shown in Fig. 7(i) and (e), the changed and unchanged classes are overlapped strongly, leading to large numbers of FP and FN. For CA_AE, some unchanged areas with large values in DI are misclassified into the changed ones. The proposed method gets the best CD results with the smallest number of FP. Compared with other CD methods, the unchanged areas can be matched exactly due to the effectiveness of the dense features learned by MC-CSSL. Therefore, it is observed that most of the unchanged areas are suppressed from Fig. 7(k). The proposed CD method generates a separable DI with an AUC value of 0.9848.

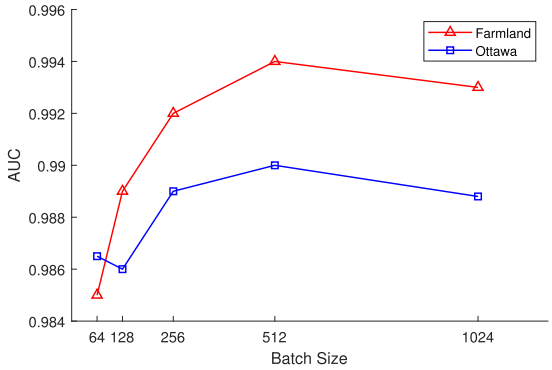


Fig. 9. With different numbers of the batch size used in dense features pretraining, AUC values are shown for the Farmland and Ottawa datasets.

D. Discussion

1) Dense Features by MC-CSSL: In MC-CSSL, the negative pairs are sampled from the same batch. When the batch size is larger, there are more negative samples for each local patch. Therefore, the selection of batch size has a strong influence on the performance of the learned features according to [39]. To evaluate the performance of MC-CSSL with different batch sizes, the dense features extracted from two temporal images with the same imaging modality are compared. As a result, an initial DI is generated based on the L2 distance between them. The AUC values of the initial DIs can assess the pretrained dense features applied to the CD task.

Fig. 9 shows AUC values for the Farmland and Ottawa datasets when different batch sizes are used in MC-CSSL. It is observed that the AUC value increases with a large batch size. When the batch size is equal to 512, the highest AUC values are achieved on both the Farmland and Ottawa datasets. However, AUC shows a little decrease when the batch size is set to 1024. The reason may be that there exists some negative pairs with similar appearance when the batch size is too large. Such negative pairs are forced to be distant in the embedding space, which is the impediment to dense feature learning.

2) Self-Similarity Matrices: As mentioned in Section IV, the initial unchanged pixels can help the feature mapping and matching. In heterogeneous cases, the truncated affinity matrices are compared to generate a distance map that is also evaluated by AUC. A higher AUC value indicates a high-quality distance map with strong separability.

In Table VI, the AUC values are listed for the distance maps by comparing self-similarity matrices on the River and Shuguang datasets. The "downsampled" version means that

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TABLE VI
AUC VALUES BY COMPARING AFFINITY OR SELF-SIMILARITY MATRICES

Data set	affinity matrices	self-similarity matrices	
		(downsampled)	(centered)
River	0.6043	0.6118	**0.6577**
Shuguang	0.9291	0.9311	**0.9428**

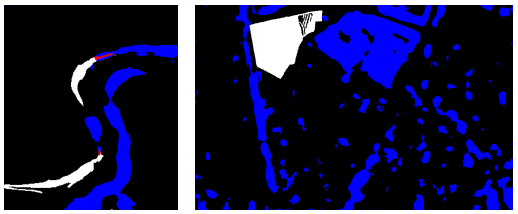


Fig. 10. Based on self-similarity matrices (centered), initial unchanged pixels are selected and marked by blue. (a) River dataset with $k_1 = 20$ and $M_I = 10^4$. (b) Shuguang dataset with $k_1 = 20$ and $M_I = 10^5$.

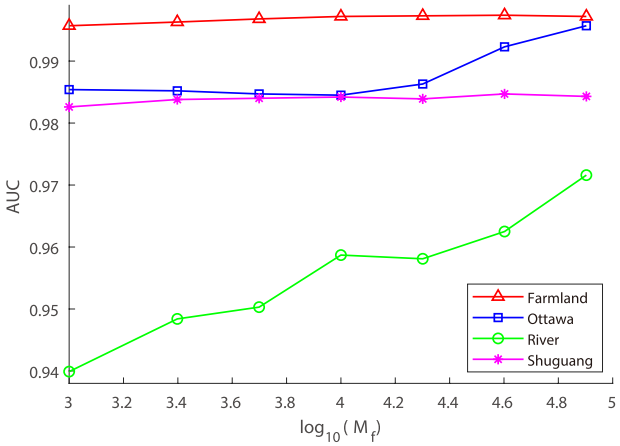


Fig. 11. AUC values of the proposed method with different numbers of unchanged pixels, that is, M_f.

q^X and q^Y are generated from p^X and p^Y with downsampling operators, respectively. Compared with affinity matrices, the "downsampled" version of truncated affinity matrices gets higher AUC values. In the "centered" version, q^X and q^Y are cropped in the center of p^X and p^Y. From Table VI, it can be seen that the "centered" version of self-similarity matrices performs best. In other words, a pair of unchanged pixels with different imaging modalities can be matched. In addition, two versions of the truncated affinity matrices also need less computation time than affinity matrices.

In Fig. 10, some initial unchanged pixels are selected from the distance map based on the "centered" version of self-similarity matrices. In Fig. 10(a), 10^4 unchanged pixels are marked by blue in the River dataset. For Shuguang dataset, 10^5 unchanged pixels are shown in Fig. 10(b).

3) Feature Mapping and Matching: In (6), $\mathcal{D}_u$ contains the unchanged pixels that are used in dense feature mapping and matching process. Let M_f denote the number of unchanged pixels when the networks stop training. To evaluate the influence of M_f on the feature mapping and matching process, the AUC value of the DI generated by (7) is also provided. Fig. 11 summarizes how AUC values vary with different numbers of M_f on all datasets. For Farmland and Shuguang datasets, $M_f = 10^3$ unchanged pixels are enough to map the dense features from two input images into the same space. When M_f increases, the DI is improved slightly. For the River dataset, the DI is not satisfied with a small number M_f of unchanged pixels. When M_f is large, the AUC value is also increased. To derive a satisfied DI, M_f is equal to 4×10^4 for Farmland and Shuguang datasets while it is 8×10^4 for Ottawa and River datasets. From Fig. 11, it can also be seen that the AUC values for the River and Shuguang datasets are always smaller than that for the Farmland and Ottawa sets. It is demonstrated that the CD task in heterogeneous cases is more challenging than that in homogeneous cases.

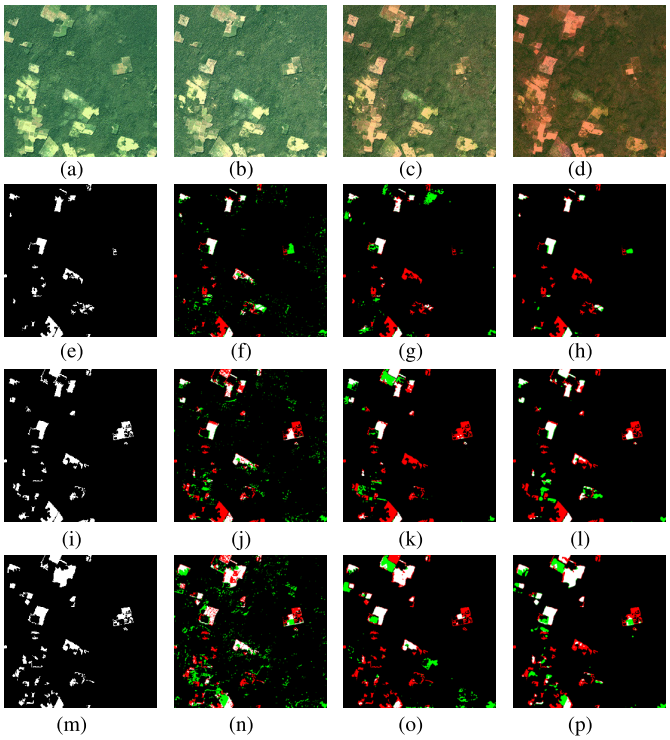


Fig. 12. False-color composite of the DEN-16N dataset and the corresponding CD results. (a) Image acquired at t_1 (January 2019). (b) Image acquired at t_2 (February 2019). (c) Image acquired at t_3 (March 2019). (d) Image acquired at t_4 (April 2019). (e) Reference image between t_1 and t_2. The confusion maps by (f) AM_HPT, (g) CA_AE, and (h) proposed between t_1 and t_2. (i) Reference image between t_1 and t_3. The confusion maps by (j) AM_HPT, (k) CA_AE, and (l) proposed between t_1 and t_3. (m) Reference image between t_1 and t_4. The confusion maps by (n) AM_HPT, (o) CA_AE, and (p) proposed between t_1 and t_4. In the confusion maps, white: TP, black: TN, green: FP, Red: FN.

E. Multitemporal Change Detection

The DEN-16N dataset from DynamicEarthNet [57] consists of four monthly images from January 2019 to April 2019. Each image has four spectral bands (blue, green, red, and NIR), as shown in Fig. 12(a)–(d). The image size is 1024×1024 and the spatial resolution is 3 m. There are two classes of land-coverings contained in the DEN-16N dataset, which are forest (or other vegetation) and soil. The binary change maps shown in Fig. 12(e), (i), and (m) were generated by comparing the semantic maps, providing a view of the forest and other vegetation changes.

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TABLE VII
CD RESULTS ON THE DEN-16N DATASET

Methods	KC	PC(%)	OE	AUC
AM_HPT	**0.4938**	96.76	8490	0.8555
	0.4588	**95.07**	**12919**	0.8717
	0.4383	92.91	18587	0.8859
CA_AE	0.4357	96.75	8525	0.8808
	0.4078	94.94	13257	0.9238
	0.4841	94.57	14246	**0.9409**
Proposed	0.4882	**97.03**	**7778**	**0.9188**
	0.5188	94.71	13857	**0.9544**
	0.4972	**95.66**	**11372**	0.9318

For the DEN-16N dataset, the proposed method is compared with AM_HPT and CA_AE since they are general CD methods on various remote sensing data. Table VII lists the corresponding CD results between t_1 and t_2, between t_1 and t_3, and between t_1 and t_4. It can be seen that the proposed CD method achieves the largest or comparable AUC values with AM_HPT and CA_AE. Therefore, the DI is separable between changed and unchanged classes after the dense features extracted by MC-CSSL are matched. In Fig. 12, the confusion maps are visualized for the DEN-16 dataset. Compared with AM_HPT and CA_AE, the proposed CD method yields less false alarms and isolated noises. The spatial correlation is guaranteed in the semantic cue of MC-CSSL, which can reduce the noise in CD maps.

VI. CONCLUSION

In this article, we propose MC-CSSL to learn dense features for CD in both homogeneous and heterogeneous cases. To model the correlation between patches, more cues are used from both the semantic and temporal views. As a result, dense features are learned efficiently, and pixels from the same class of land-covering are encouraged to be close. In addition, the local self-similarity is introduced to extract change information for two temporal patches though with different appearances and statistics. The distance between two truncated affinity matrices can help select some initial unchanged pixels. These unchanged pixels serve as the supervision to map the dense features into a common space. After comparing the dense features in the common space, a DI can be generated to derive more unchanged pixels. The proposed CD method is unsupervised and performs well in both homogeneous and heterogeneous cases.

The proposed method also has some limitations. For example, the input images are segmented into several classes to provide a semantic cue for dense features pretraining. However, it is difficult to get a useful classification map for the complicated input image with more types of land-coverings. In future work, we will exploit a good solution to provide semantic cues for complicated input images.

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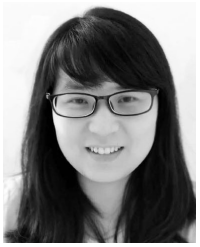
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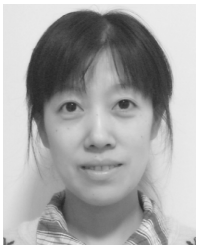


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